BTS–III–11.18–1724 Reg. No. [ ]

B

B.Tech. Degree III Semester Examination November 2018

ME 15-1304 FLUID MECHANICS
(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A
(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) Explain surface tension and capillarity.
(b) State and prove pascals law.
(c) Explain single column manometer. Obtain the equation for pressure measurement using it.
(d) Explain steady, unsteady uniform and non uniform flow field.
(e) Explain Venturi meter, Orifice meter and Flow nozzle meter.
(f) Explain minor losses pipe flow.
(g) Define and explain stream function and velocity potential.
(h) Explain source, sink and doublet.
(i) Explain formation of boundary layer on a flat plate.
(j) Explain separation of boundary layer.

PART B

$(4 \times 10 = 40)$

II. (a) The dynamic viscosity of an oil used for lubrication between a shaft and sleeve is 6 poise. The shaft is of diameter 0.4 m and rotates at 190 rpm. Calculate the power lost in bearing for a sleeve length of 90 mm . The thickness of oil film is 1.5 mm. (4)
(b) Explain experimental method of determination of metacentric height of a floating body. (2)
(c) A solid cylinder of diameter 4.0 m has a height of 4 m. Find the metacentric height of the cylinder if the specific gravity of the material of the cylinder is 0.6 and it is floating in water with its axis vertical. State whether the equilibrium is stable or unstable. (4)

OR

III. (a) Explain micro manometer. Obtain the equation of pressure measurement using it. (3)
(b) The left leg of a U-tube mercury manometer is connected to a pipeline conveying water, the level of mercury in the leg being 0.8 m below the centre of pipe line and the right leg is open to atmosphere. The level of mercury in the right leg is 0.65 m above that in the left leg and the space above mercury in the right leg contains Benzene of specific gravity 0.88 to a height of 0.3 m. Find the pressure in the pipe. (4)
(c) Find the magnitude and direction of the resultant force due to water acting on a roller gate of cylindrical form of 4 m diameter when the gate is placed on the dam in such a way that water is just going to spill. Take the length of the gate as 8 m. (3)

(P.T.O.)

IV. (a) Explain Bernoulli's equation. What are the assumptions and limitations? (2)

(b) Derive an equation for the time of emptying of a vertical cylindrical tank. (4)

(c) Derive an equation for the discharge through a venturimeter. (4)

OR

V. (a) State and prove Darcy-Weisbach equation. (5)

(b) An oil of specific gravity 0.9 and viscosity 0.06 Poise is flowing through a pipe of diameter 200 mm at the rate of 60 litres/second. Find the head lost due to friction for a 500 m length of the pipe. Find the power required to maintain this flow. (5)

VI (a) What is meant by Magnus effect? Explain. (2)

(b) Explain vorticity. Obtain the expression for vorticity in terms of angular velocity. (4)

(c) A fluid flow field is given by $V = x^2yi + y^2zj - (2xyz + yz^2)k$. Prove that it is a case of possible steady incompressible fluid flow. Calculate the velocity and acceleration at the point (2, 1, 3) (4)

OR

VII. (a) Explain flownet. (2)

(b) The velocity components in a two dimensional flow field for an incompressible flow are expressed as $u = y^3/3 + 2x - x^2y$, $v = xy^2 - 2y - x^3/3$. (i) Show that these functions represent a possible case of an irrotational flow. (ii) Obtain the expression for the steam function (iii) Obtain the expression for the velocity potential (8)

VIII. (a) Explain displacement thickness, momentum thickness and energy thickness. Derive the relations for each of these in terms of local and free stream velocity. (6)

(b) Find the displacement thickness, the momentum thickness and energy thickness for the velocity distribution in the boundary layer given by $\frac{u}{U} = 2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2$. (4)

OR

IX. (a) Derive Von Karman momentum integral equation. (6)

(b) A thin plate is moving in still atmospheric air at a velocity of 5 m/s. The length of the plate is 0.6 m and width 0.5 m. Calculate: (i) thickness of boundary layer at the end of the plate (ii) drag force on one side of the plate. Take density of air as 1.24 kg/m^3 and kinematic viscosity 0.15 stokes. Take $\delta = \frac{4.91x}{\sqrt{Re_x}}$. (4)

***